

Network Troubleshooting USING PACKET SNIFFER WIRESHARK

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In industries, many systems are based on networks comprising different devices and controllers that work on some protocols. Often the need arises to monitor the entire communication process to make the system more efficient. Sometimes, there is a need to monitor the process to remove numerous faults and errors. A packet sniffer is just the software needed to meet such requirements. This article describes how you can check and analyse data passing through a network using a popular packet sniffer called Wireshark.

What a packet sniffer is

Packet sniffers or protocol analysers are tools that are commonly used to diagnose network-related problems. These can also be used by hackers for spying on network user traffic and collecting passwords.

In its simple form a packet sniffer simply captures all packets of data that pass through a given network interface. Typically, it only captures packets that were intended for the machine in question. However, if placed into promiscuous mode, it is also capable of capturing all packets traversing the network, regardless of destination.

How it works

Packet sniffers work by intercepting and logging network traffic that these can see via the wired or wireless network interface to the computer in use.

On a wired network, what can be captured depends on the structure of the network. A packet sniffer might be able to see traffic on an entire

	AIX	FreeBSD	HP-UX	Irix	Linux	Mac OS X	NetBSD	OpenBSD	Solaris	Tru64 UNIX	Windows
Physical Interfaces											
ATM	Unknown	Unknown	Unknown	Unknown	Yes	No	Unknown	Unknown	Yes	Unknown	Unknown
Bluetooth	No	No	No	No	Yes	No	No	No	No	No	No
CiscoSDLC	Unknown	Yes	Unknown	Unknown	Yes	Unknown	Yes	Yes	Unknown	Unknown	Unknown
Ethernet	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FDDI	Unknown	Unknown	Unknown	Unknown	Yes	No	Unknown	Unknown	Yes	Unknown	Unknown
FrameRelay	Unknown	Unknown	No	No	Yes	No	Unknown	Unknown	No	No	No
IrDA	No	No	No	No	Yes	No	No	No	No	No	No
PPP	Unknown	Unknown	Unknown	Unknown	Yes	Yes	Unknown	Unknown	No	Unknown	Yes
TokamiRing	Yes	Yes	Unknown	No	Yes	No	Yes	Yes	Yes	Unknown	Yes
USB	No	No	No	No	Yes	No	No	No	No	No	No
WLAN	Unknown	Yes	Unknown	Unknown	Yes	Yes	Yes	Yes	Unknown	Unknown	Yes
Virtual Interfaces											
Loopback	Unknown	Yes	No	Unknown	Yes	Yes	Yes	Yes	No	Yes	N/A
VLAN Tags	Yes	Yes	Yes	Unknown	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Fig. 1: Network types supported on various platforms by Wireshark

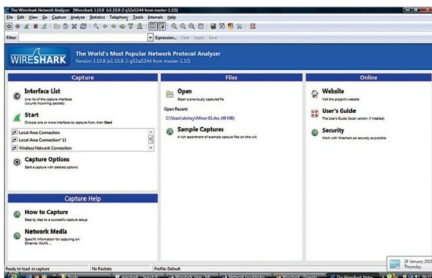


Fig. 2: Home screen for Wireshark for Windows 32-bit

network or only a certain segment of it, depending on how network switches are configured, placed, etc.

On a wireless network, a packet sniffer can usually only capture one channel at a time, unless the host computer has multiple wireless interfaces that allow for multichannel capture.

Once raw packet data is captured, packet sniffing software must analyse it and present it in human-readable form so that the person using the software can make sense of it. The person analysing the data can view details of the conversations happening between two or more nodes on the network.